

WLOG

let x be $\frac{\pi}{2} - x$.

$$\cos\left(\frac{\pi}{2} - x - \beta\right) = \underbrace{\cos\left(\frac{\pi}{2} - x\right)\cos\beta}_{=\sin x} + \underbrace{\sin\left(\frac{\pi}{2} - x\right)\sin\beta}_{\cos x}$$

$$= \sin(x+\beta) = \sin x \cos \beta + \cos x \sin \beta.$$

$$\therefore \sin(x+\beta) = \sin x \cos \beta + \cos x \sin \beta //$$

$$\sin(x-\beta) = \sin x \cos \beta - \cos x \sin \beta //$$

$$\left(\tan \theta = \frac{\sin \theta}{\cos \theta} \right) \Rightarrow \frac{\sin(x+\beta)}{\cos(x+\beta)} = \frac{\sin x \cos \beta - \cos x \sin \beta}{\cos x \cos \beta - \sin x \sin \beta}$$

$\theta = x+\beta$

$$\therefore \tan(x+\beta) = \frac{\sin x \cos \beta - \cos x \sin \beta}{\cos x \cos \beta - \sin x \sin \beta} \Rightarrow \tan(x+\beta) = \frac{\tan x + \tan \beta}{1 - \tan x \tan \beta}$$

let β be $-\beta$.

$$\Rightarrow \tan(-\beta) = -\tan \beta.$$

$$\Rightarrow \tan(x-\beta) = \frac{\tan x + \tan(-\beta)}{1 - \tan x \tan(-\beta)}$$

$$= \frac{\tan x - \tan \beta}{1 + \tan x \tan \beta}.$$

$$\therefore \tan(x-\beta) = \frac{\tan x - \tan \beta}{1 + \tan x \tan \beta}$$